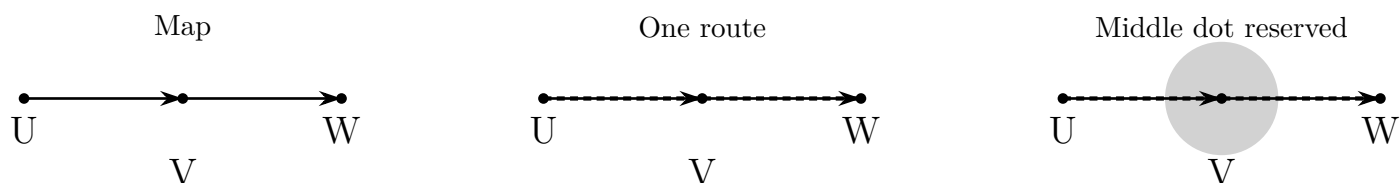
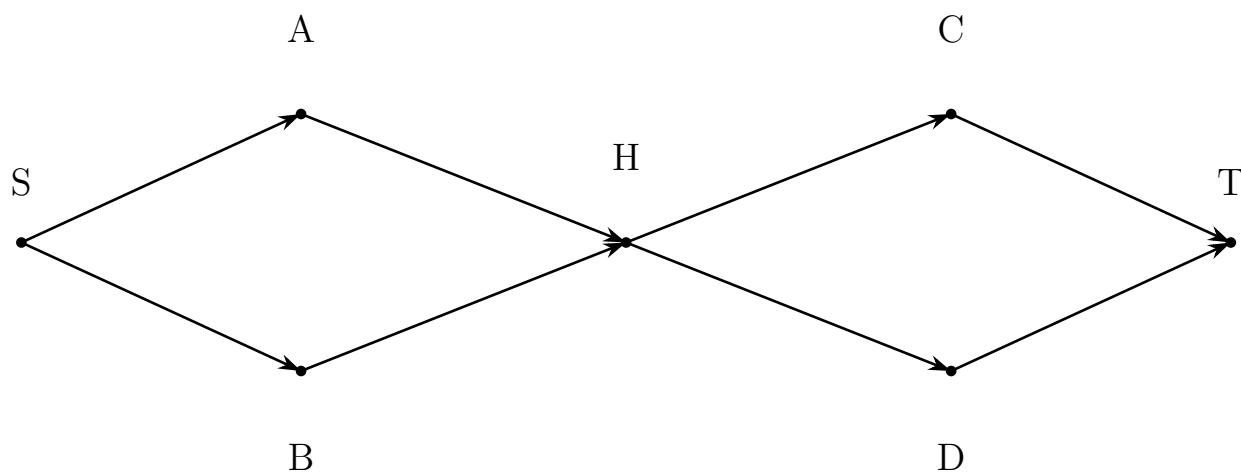


A route follows the arrows. Keep all routes in one attempt drawn at once; no two routes may use the same arrow. Use solid and dashed lines.

For one route per middle dot, leave a marker on every middle dot the route uses.

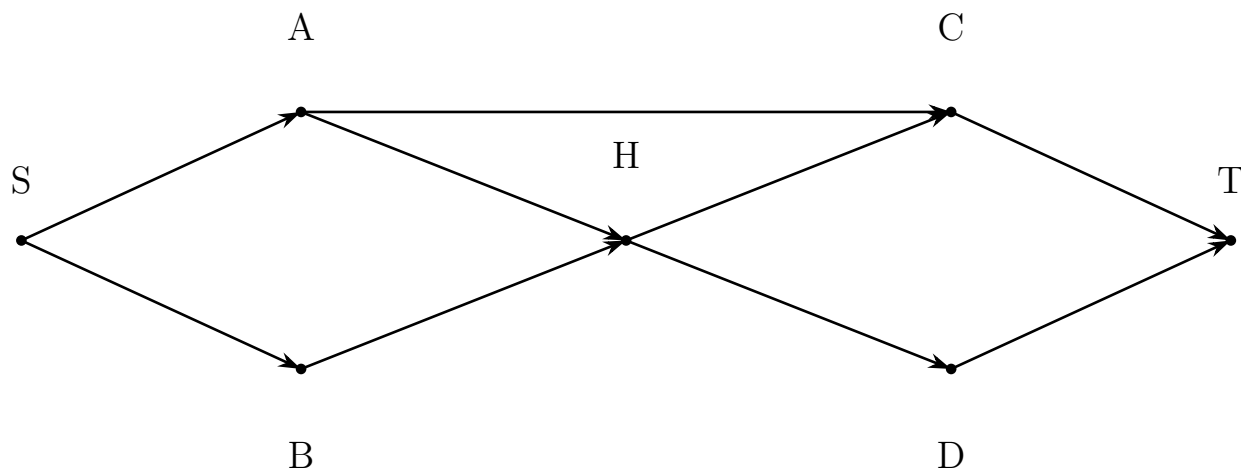


Problem 1: Fit as many S-to-T routes as you can on each map. Try sharing middle dots, then erase the routes and remove the markers before trying one route per middle dot.



Shared middle dots: ____

One route per middle dot: ____



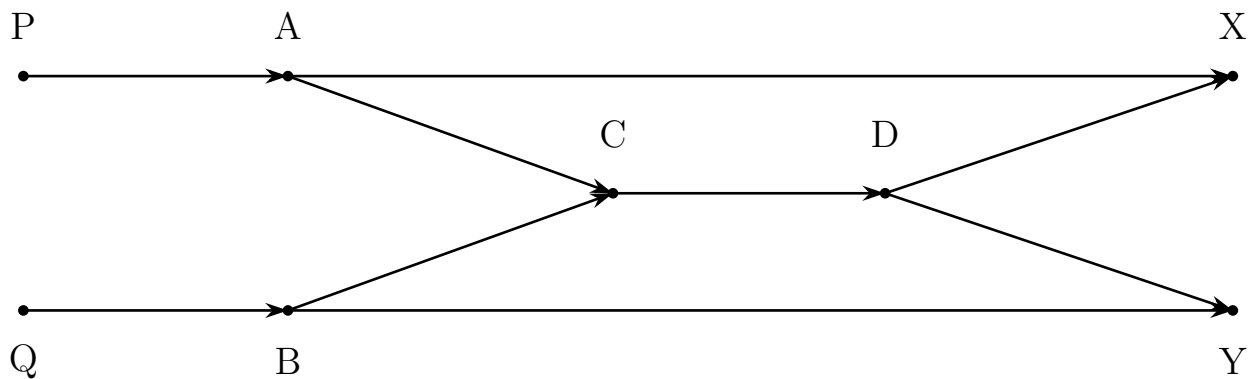
Shared middle dots: ____

One route per middle dot: ____

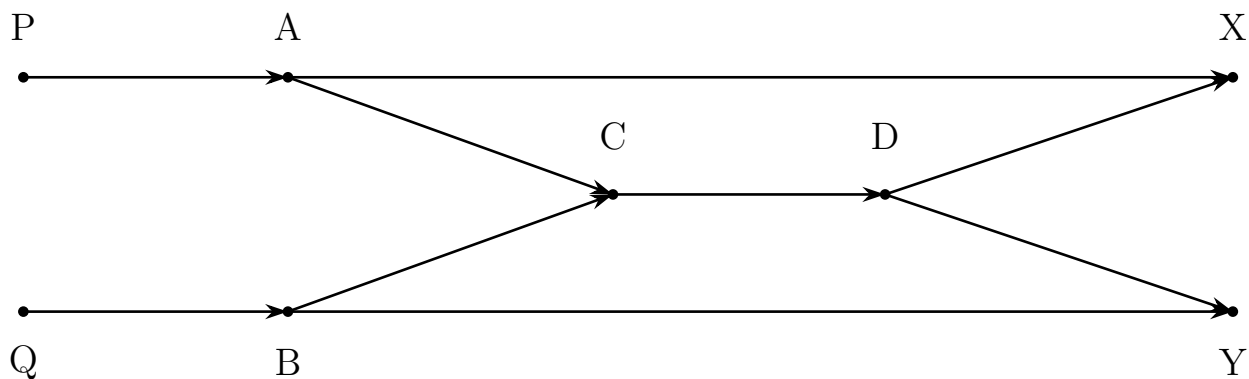
Keep both routes drawn at once. They follow the arrows and may share a dot, but cannot share an arrow. Use solid and dashed lines.

Problem 2: Which pairings can fit at the same time? On the last map, choose two different finishes for P and Q so that both routes fit.

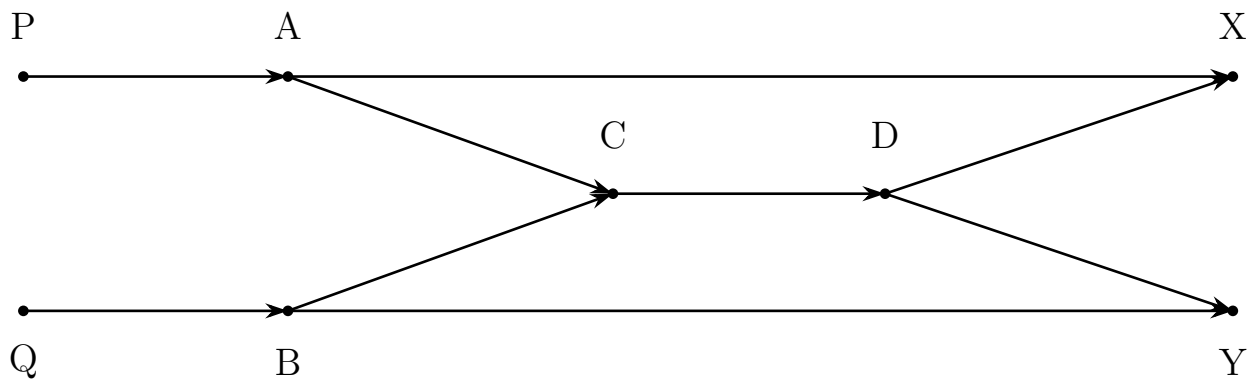
P $\longrightarrow$ X Q $\longrightarrow$ Y Fits / Impossible



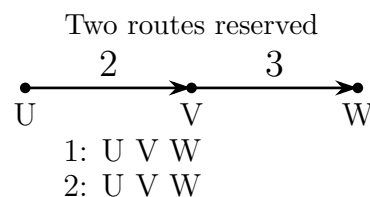
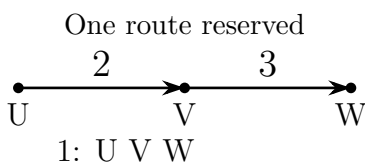
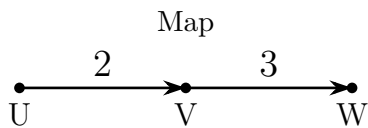
P $\longrightarrow$ Y Q $\longrightarrow$ X Fits / Impossible



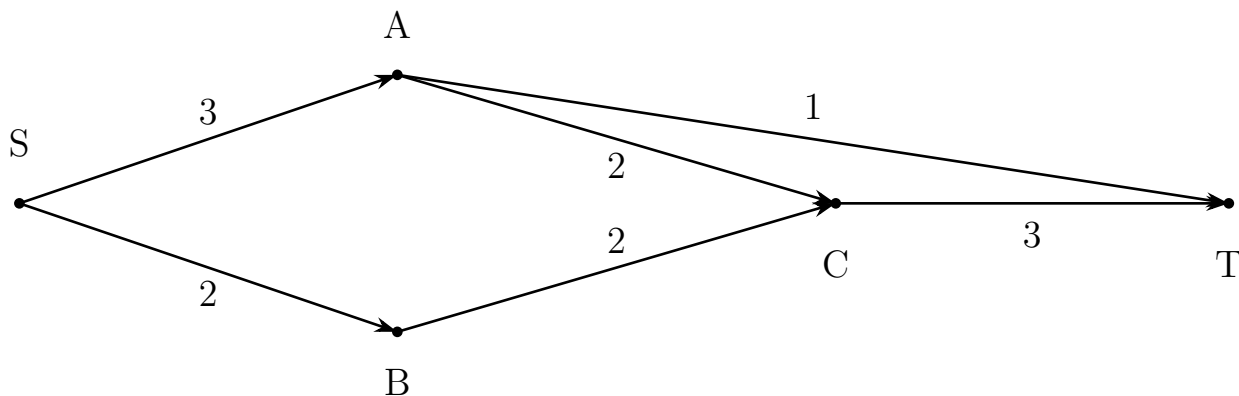
P $\longrightarrow$ _____ Q $\longrightarrow$ _____



Each row reserves one whole route. A number beside an arrow limits how many reserved routes may use it. Keep all rows for one attempt; repeated routes are allowed.



Problem 3: Fit the most S-to-T routes on each map. Cross out arrows that stop every S-to-T route, with numbers adding to your route count.



1: _____

4: _____

2: _____

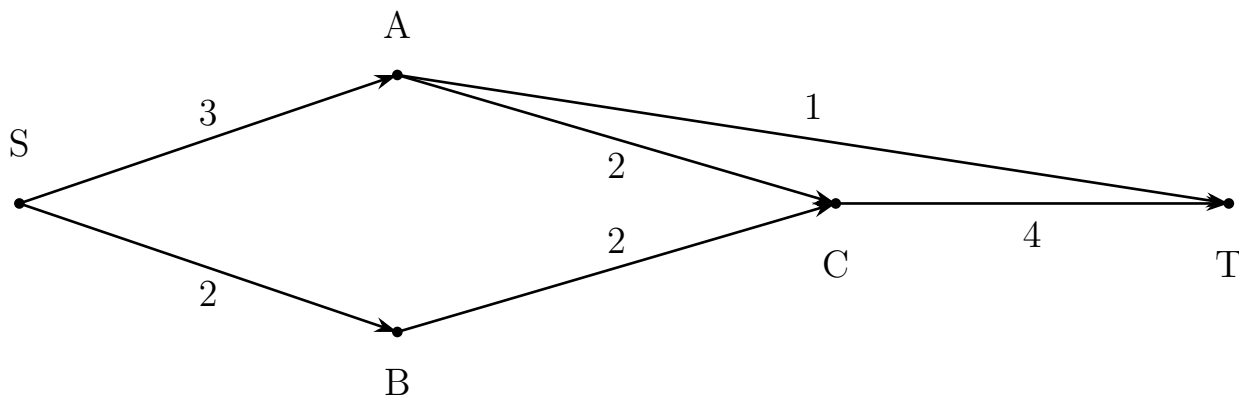
5: _____

3: _____

6: _____

Routes: _____

Crossed-out numbers add to: _____



1: _____

4: _____

2: _____

5: _____

3: _____

6: _____

Routes: _____

Crossed-out numbers add to: _____